

Martim Vicente

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Education

Instituto Superior Técnico

BSc in Computer Science and Engineering

Lisbon, Portugal

Sep. 2024 – Jun. 2027

- **Average:** 15.75 / 20.0
- **Honors:** Técnico ALUMNI Scholarship, Gulbenkian Merit Scholarship, WARD Analytics Scholarship

Experience

Software Team Lead, Rocket Experiment Division – Lisbon, PT

May 2026 – Present

- Led the software team through 3 testing campaigns and the EuRoC competition, owning system architecture, integration, validation, and end-to-end software readiness.
- Developed safety-critical firmware for ignition, avionics and filling systems on STM32/FreeRTOS, implementing sensor-fusion filters and fail-safe mechanisms for deterministic fault recovery.
- Refactored STM32 peripheral drivers (IMU, GPS, Valves, Sensors) into a modular Hardware Abstraction Layer (HAL) in C, cutting new-sensor integration time by 60% and enabling mock-based unit tests.
- Rebuilt Ground Support Equipment (GSE) backend in async Python, reducing production LOC by 47% while maintaining full feature parity.
- Architected a hardware-agnostic telemetry simulator enabling full end-to-end Ground Support Equipment bench validation, eliminating all hardware dependencies for testing.
- Migrated all systems from a proprietary protocol to MAVLink v2, unifying telemetry parsing, reducing development, integration and debugging time.

Embedded Software Engineer, Rocket Experiment Division – Lisbon, PT

Sep. 2025 – May 2026

- Developed the foundational firmware for the filling system on STM32/FreeRTOS, implementing the initial event-driven control logic for valve actuation, pressure monitoring, and timeout-based abort sequences.
- Built an internal platform centralizing architecture decisions, test procedures, and onboarding guides, establishing a single source of truth, streamlining team workflows.

Software Engineer, Técnico Investment Club – Lisbon, PT

Sep. 2025 – May 2026

- Re-architected the Python trading runtime into decoupled ingestion and strategy services, slashing LOC by 80%, achieving 100% live stability, and cutting new strategy implementation time by 50%.
- Designed the architecture for a low-latency C++ execution engine targeting sub-50µs order-to-fill, centered on non-blocking pipelines, zero-copy data paths, and fast-fail validation components.

Projects

Python Trading Engine

GitHub

- Built a containerized algorithmic trading platform with decoupled ETL ingestion and execution services orchestrated via Docker Compose.
- Designed an interface-driven architecture (IStrategy, IProvider) and a factory-pattern strategy loader, allowing new algorithms to be deployed via YAML configurations without touching core logic.
- Integrated exchange adapters for Alpaca and Binance through a pluggable provider interface, enabling seamless switching between live data sources.

Skills

Programming: C++, C, Python, Go, Java

Tools: Linux, Git, CMake, Make, Docker, QEMU

Embedded: STM32, ESP32, FreeRTOS, CAN, I2C, UART, SPI, ADC, MQTT, MAVLink

Testing: Pytest, GoogleTest, GitHub Actions

Languages: Portuguese (Native), English (Professional)